

केन्द्रीय विद्यालय संगठन, कोलकाता संभाग

KENDRIYA VIDYALAYA SANGATHAN, KOLKATA REGION

प्री-बोर्ड परीक्षा / PRE-BOARD EXAMINATION – 2025-26

कक्षा/CLASS – X

विषय/SUB. – Artificial Intelligence (417)

अधिकतम अंक/MAX. MARKS – 50

समय/TIME – 02 घंटे/Hours

MARKING SCHEME

SECTION A: OBJECTIVE TYPE QUESTIONS

1	Answer any 4 out of the given 6 questions on Employability Skills(4 x 1 = 4)	
i.	c) Financial barrier	1
ii.	b) Measurable	1
iii.	a) Self-motivation	1
iv.	c.) Operating System	1
v.	d) All of the above	1
vi.	a) Taking care of future generations.	1
2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	b) AI Bias	1
ii.	a) Both Assertion and Reasoning are true, and Reasoning is the correct explanation of the Assertion.	1
iii.	a) Supervised Learning	1
iv.	_____ Evaluation _____	1
v.	c) Image Processing; Computer Vision	1
vi.	b.) 1280 pixels across width and 1024 pixels in height	1
3.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	b) Data Acquisition	1
ii.	b) Label	1
iii.	c) 96%	1
iv.	(a) Both A and R are true and R is the correct explanation for A	1
v.	a) Facial Recognition	1
vi.	13	1
4.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	a) Step-by-step guidance	1
ii.	a) Both A & R are correct.	1
iii.	a) Overfitting.	1
iv.	True	1
v.	c) 0.5	1
vi.	a) Instance Segmentation	1
5.	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	c) Can be less efficient for large datasets.	1
ii.	d) 0.75	1
iii.	d) 0 to 255	1
iv.	d) True Negative	1
v.	b) Smart Bot	1
vi.	stemming	1

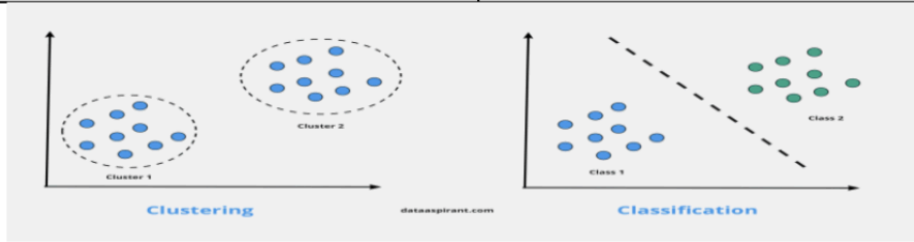
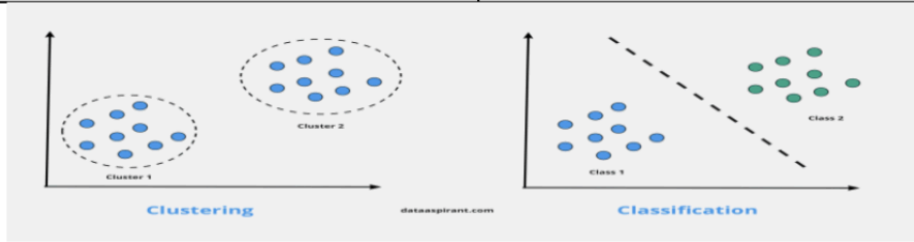
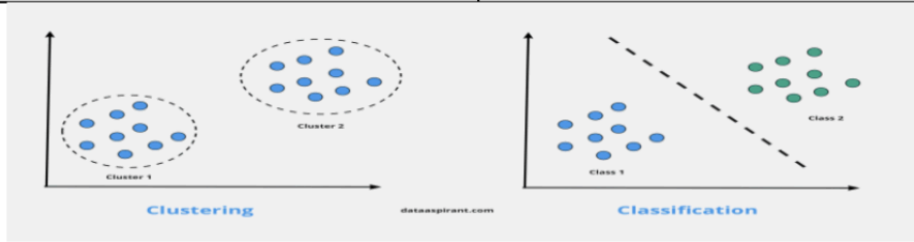
SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words.

6.	• Smiling when meeting someone. • Keeping the face relaxed. • Matching expressions with your words. • Nodding while listening. (Any two Points)	2															
7.	Some of stress management techniques are- Yoga, Physical Exercise, Meditation, Nature Walks, Going on Vacations, Diet and Sleep, Positive Thinking, Listening to good music, Laughing aloud etc. (Any four- 0.5 marks for each)	2															
8.	A folder is a container to carry files. A folder can have sub folders as well. Following are the important differences between File and Folder. <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <thead> <tr> <th style="text-align: left;">Aspect</th><th style="text-align: left;">File</th><th style="text-align: left;">Folder</th></tr> </thead> <tbody> <tr> <td>1. Definition</td><td>A file is a collection of data or information stored on a computer.</td><td>A folder is a container used to organize and store files (and sometimes other folders).</td></tr> <tr> <td>2. Content</td><td>It usually contains text, images, videos, programs, or other data.</td><td>It contains files and/or subfolders, but not raw data itself.</td></tr> <tr> <td>3. Extension</td><td>Files have extensions like <code>.txt</code>, <code>.jpg</code>, <code>.docx</code>, <code>.mp3</code> etc.</td><td>Folders do not have extensions.</td></tr> <tr> <td>4. Representation (Icon)</td><td>Represented by icons like a sheet, image, or document.</td><td>Represented by a yellow folder icon (by default in most systems).</td></tr> </tbody> </table> (Any four- 0.5 marks for each correct difference.)	Aspect	File	Folder	1. Definition	A file is a collection of data or information stored on a computer.	A folder is a container used to organize and store files (and sometimes other folders).	2. Content	It usually contains text, images, videos, programs, or other data.	It contains files and/or subfolders, but not raw data itself.	3. Extension	Files have extensions like <code>.txt</code> , <code>.jpg</code> , <code>.docx</code> , <code>.mp3</code> etc.	Folders do not have extensions.	4. Representation (Icon)	Represented by icons like a sheet, image, or document.	Represented by a yellow folder icon (by default in most systems).	2
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9.	An entrepreneur helps the society in the following ways: (i) Welfare programs: Business persons financially contribute to many programs, like helping the blind, orphans, widows, organizing donation camps, etc. They also donate items such as blankets, clothes, medicines etc in times of crisis or disasters. (ii) Charitable institutions: Various educational, medical and vocational training institutions are set-up by business people to provide less- privileged with benefits which they normally cannot afford. (iii) Donations: Businessmen donate a lot of money to charity, which is used for the welfare of the downtrodden and poor people and helps them in improving their living conditions. (iv) Sponsorship: Some business persons also sponsor candidates for higher education or fund children in orphanages. (v) Advisors to respective governments: Many successful business persons also participate in government activities to promote the well-being of the citizens. The government often seeks their advice on certain social issues and asks for their participation in economic activities as a part of CSR (Corporate Social Responsibility). (Any two - 1 marks for each)	2															
10.	Organic farming is where farmers do not use chemical pesticides and fertilisers to increase their production. Benefits: • It helps in growing better quality, chemical-free crops. • It maintains the soil quality for future use. (1 mark for definition; ½ mark for each benefit)	2															

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

11.	Principles of bioethics: • Respect for Autonomy. • Do not harm. • Ensure maximum benefit for all. • Give justice. (01 marks for each)	2																		
12.	Case 1: Predicting whether a customer is eligible for a bank loan or not? - Classification Case 2: Predicting weather for next 24 hours - Regression	2																		
13.	<table><tr><th>Parameter</th><th>CLASSIFICATION</th><th>CLUSTERING</th></tr><tr><td>Basic</td><td>process of classifying the input instances based on their corresponding class labels</td><td>grouping the instances based on their similarity without the help of class labels</td></tr><tr><td>Need</td><td>it has labels so there is need of training and testing dataset for verifying the model created</td><td>there is no need of training and testing dataset</td></tr><tr><td>Complexity</td><td>more complex as compared to clustering</td><td>less complex as compared to classification</td></tr><tr><td>Example Algorithms</td><td>Logistic regression, Naive Bayes classifier, Support vector machines, etc.</td><td>k-means clustering algorithm, Fuzzy c-means clustering algorithm, Gaussian (EM) clustering algorithm, etc.</td></tr><tr><td></td><td colspan="2"></td></tr></table> (Any two correct difference - 01 marks for each)	Parameter	CLASSIFICATION	CLUSTERING	Basic	process of classifying the input instances based on their corresponding class labels	grouping the instances based on their similarity without the help of class labels	Need	it has labels so there is need of training and testing dataset for verifying the model created	there is no need of training and testing dataset	Complexity	more complex as compared to clustering	less complex as compared to classification	Example Algorithms	Logistic regression, Naive Bayes classifier, Support vector machines, etc.	k-means clustering algorithm, Fuzzy c-means clustering algorithm, Gaussian (EM) clustering algorithm, etc.				2
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14.	<table><tr><td>100 (TP)</td><td>62 (FP)</td></tr><tr><td>290 (FN)</td><td>47 (TN)</td></tr></table> (0.5 marks for each correct placement of TP/TN/FP/FN)	100 (TP)	62 (FP)	290 (FN)	47 (TN)	2														
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15.	The most common pixel format is the byte image, where each pixel value is stored as an 8-bit integer, giving a range of possible values from 0 to 255. In this format, 0 is typically taken as no colour or black, and 255 is taken as full colour or white. In computer systems, data is represented in the binary system, using ones and zeros. Each bit can have two possible values, and since 1 byte = 8 bits, this results in 256 possible values, ranging from 0 to 255. Therefore, each pixel in a byte image uses 1 byte of data. (1 mark for correctly stating the definition; 1 mark for explaining how the binary system relates to this format)	2																		

16.			2
		Stemming	
	happily	happi	
	caring	car	
(0.5 marks for each)			

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

17.	a) Two reasons why the algorithm gave biased results: • The algorithm unfairly penalized students who are not fluent in English, assuming slower form completion indicates lower competence, when it may simply reflect language challenges. • Students without access to a computer or stable internet at home took longer to complete the application, but this delay is due to external socio- economic factors, not lack of ability or confidence. (1 mark for each valid reason) b) Two bioethics principles that can solve this problem are: • Do Not Harm: The algorithm unintentionally harms students from less privileged backgrounds by reducing their chances of selection. To follow this principle, the AI system should be redesigned to avoid causing unfair disadvantages. • Give Justice: Justice requires that all students are treated fairly and equitably. The system should not discriminate based on factors like language fluency or digital access. Incorporating diverse data and adjusting evaluation criteria would help ensure equal opportunity for all applicants. (½ mark for each correct bioethics principle and ½ mark for the each relevant explanation)	4
18.	A. Classification B. Regression C. Clustering D. Association	4
19.	A. (½ mark each for correctly identifying and writing values of TP, FP, TN,FN) B. Classification Accuracy $\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$ $= \frac{TP + TN}{TP + FP + TN + FN}$ $= \frac{40 + 30}{40 + 20 + 30 + 10}$ $= \frac{70}{100} = \underline{\underline{0.7}}$ Accuracy% = 70% (½ mark for formula; ½ mark for the answer) C. Wrong Predictions = FP+FN =20+10 =30 (½ mark for formula; ½ mark for the answer)	4
20.	A. Anomaly Detection B. Object Classification C. Object Identification D. Digit Recognition (1 mark each)	4

21.											4
	Document	AI	students	love	studying	is	transforming	education	teachers	and	explore
	Doc 1	1	1	1	1	0	0	0	0	0	0
	Doc 2	1	0	0	0	1	1	1	0	0	0
	Doc 3	1	1	0	0	0	0	0	1	1	1
	(correct 04 steps and 1 marks for each steps BoW algorithm)										
